9.5.1

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Question:

Find the roots of the equation $x^2+3x-10=0$ by representing it as a 2D conic section and finding its points of intersection.

Solution:

We express the equation as a parabola $y = x^2 + 3x - 10$, which can be written in the standard conic form $\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0$ as:

$$x^2 + 3x - y - 10 = 0 \tag{1}$$

The components of the conic form are:

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad 2\mathbf{u} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \implies \mathbf{u} = \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix}, \quad f = -10 \tag{2}$$

We find the intersection with the x-axis, which has the parametric form $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$, with a point on the line $\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and direction vector $\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

The values of κ for the intersection points The solution is:

$$\kappa = \frac{-\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{(\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}))^2 - (\mathbf{m}^\top\mathbf{V}\mathbf{m})(\mathbf{h}^\top\mathbf{V}\mathbf{h} + 2\mathbf{u}^\top\mathbf{h} + f)}}{\mathbf{m}^\top\mathbf{V}\mathbf{m}} \tag{3}$$

We will now calculate each term in this formula.

Denominator Term:

$$\mathbf{m}^\top\mathbf{V}\mathbf{m} = \begin{pmatrix} 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \tag{4}$$

Numerator Term 1: Since $\mathbf{h} = \mathbf{0}$, this term is $-\mathbf{m}^\top\mathbf{u}$.

$$-\mathbf{m}^\top\mathbf{u} = -\left(\begin{pmatrix} 1 & 0 \end{pmatrix}\begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix}\right) = -\frac{3}{2} \tag{5}$$

Term under the Square Root: Since $\mathbf{h} = \mathbf{0}$, this term is $(\mathbf{m}^\top\mathbf{u})^2 - (\mathbf{m}^\top\mathbf{V}\mathbf{m})(f)$.

$$(\mathbf{m}^\top\mathbf{u})^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4} \tag{6}$$

$$(\mathbf{m}^\top\mathbf{V}\mathbf{m})(f) = (1)(-10) = -10 \tag{7}$$

$$\text{under root term} = \frac{9}{4} - (-10) = \frac{9}{4} + \frac{40}{4} = \frac{49}{4} \tag{8}$$

Now, we substitute these calculated values back into the formula for κ:

$$\kappa = \frac{-3/2 \pm \sqrt{49/4}}{1} = -\frac{3}{2} \pm \frac{7}{2} \tag{9}$$

This gives two values for κ:

$$\kappa_1 = -\frac{3}{2} + \frac{7}{2} = \frac{4}{2} = 2 \tag{10}$$

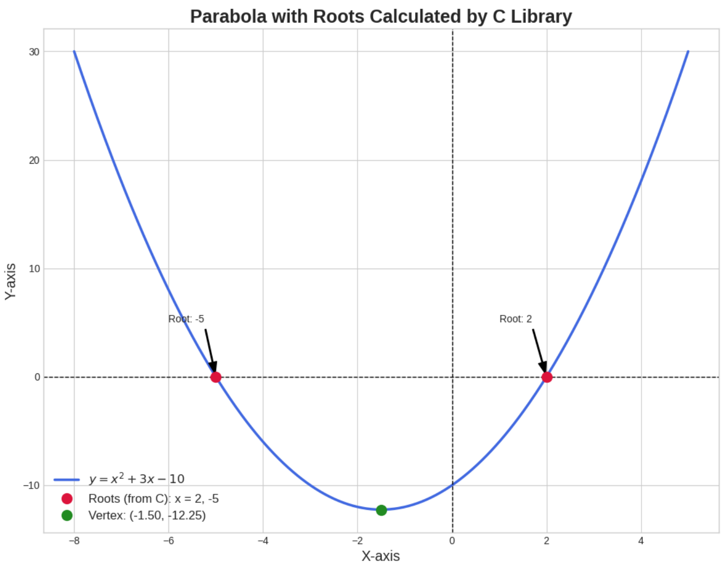
$$\kappa_2 = -\frac{3}{2} - \frac{7}{2} = -\frac{10}{2} = -5 \tag{11}$$

The points of intersection are found using $\mathbf{x_i} = \mathbf{h} + \kappa_i \mathbf{m}$:

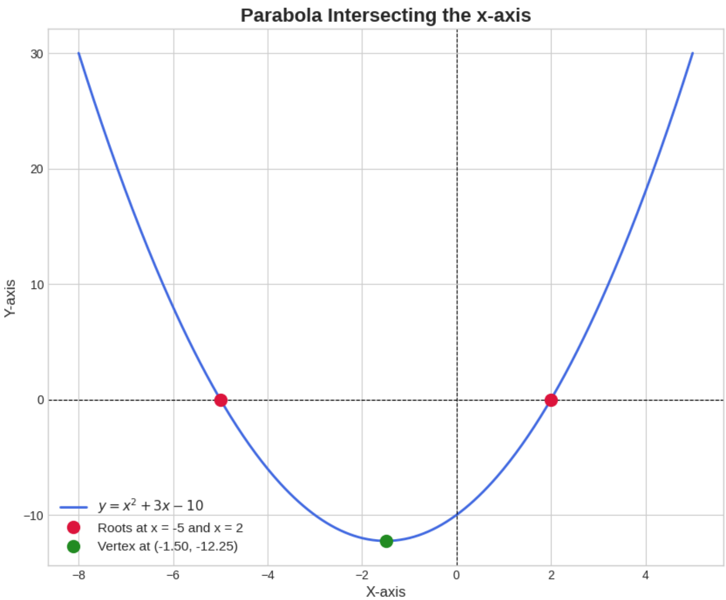
$$\mathbf{x_1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \tag{12}$$

$$\mathbf{x_2} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 5\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix} \tag{13}$$

The roots of the original equation are the x-coordinates of these intersection points.
The roots are 2 and -5.



Plot


Parabola Intersecting the x-axis
30
20
10
0
−10
Y-axis
−8
−6
−4
−2
0
2
4
X-axis
$y = x^2 + 3x - 10$
Roots at x = -5 and x = 2
Vertex at (-1.50, -12.25)

Plot